

A PROJECT TITLE
ON
Fake News Detection Using NLP, TF-IDF and BERT

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27

by
Name Kompelli Saanvi roll 2520030083
Name Yerrolla Eekshitha roll 2520030615

Under the Esteemed guidance of
Jyothsna Datti
Assistant Professor
CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Fake News Detection Using NLP, TF-IDF and BERT

ABSTRACT:

Fake news has become a major problem due to the rapid growth of social media, online news platforms, and digital communication. False or misleading information can spread quickly and may influence people's opinions, decisions, and public awareness. This project focuses on developing a machine learning and natural language processing-based system for detecting whether a news article is fake or real using Natural Language Processing (NLP), TF-IDF, Logistic Regression, and a BERT Transformer model.

A suitable fake news dataset containing news articles and their corresponding labels will be used for this project. The textual data will be pre-processed by removing unnecessary characters, handling missing values, tokenizing the text, removing stop words, and preparing the news content for feature extraction. TF-IDF (Term Frequency-Inverse Document Frequency) will be used to convert the textual data into numerical features. These features will be provided to a Logistic Regression classifier as the primary machine learning approach for classifying news articles as fake or real.

As an advanced approach, a pre-trained BERT Transformer model will also be used for fake news classification. BERT can capture contextual relationships between words and sentences and can therefore provide a deeper understanding of the news content. The performance of the TF-IDF with Logistic Regression approach and the BERT-based approach will be evaluated and compared using accuracy, precision, recall, F1-score, and confusion matrix.

The main aim of this project is to develop an effective system that can automatically classify news as fake or real based on its textual content. The comparison between the TF-IDF-based machine learning approach and the BERT-based approach will help identify their effectiveness for fake news detection. The system can serve as a supportive tool for identifying potentially misleading information and should not be considered a replacement for human fact-checking or professional verification.

INTRODUCTION:

Fake news refers to false, misleading, or fabricated information presented in the form of news. With the increasing use of social media and online platforms, news can reach a large number of people within a very short period. Although digital platforms provide fast access to information, they also make it easier for inaccurate or intentionally misleading content to spread. Therefore, automatically identifying fake news has become an important application of machine learning and Natural Language Processing.

Machine learning techniques can analyze large amounts of textual information and identify patterns that may help distinguish fake news from genuine news. In this project, NLP techniques, TF-IDF, Logistic Regression, and a BERT Transformer model are used for fake news detection.

NLP techniques will be used to clean and preprocess the textual data. TF-IDF will then be used to convert the processed news text into numerical feature vectors. Logistic Regression will use these features to classify news articles as fake or real. This approach provides a simple and efficient machine-learning solution for text classification.

As an advanced approach, BERT, a Transformer-based language model, will also be used for classification. BERT can understand the context of words by considering their surrounding words. This makes BERT useful for analyzing the meaning and context of news content.

In this project, the dataset will be cleaned and divided into training and testing data. The TF-IDF with Logistic Regression model and the BERT model will be trained and evaluated using suitable classification metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

The final objective is to compare the performance of the TF-IDF-based machine learning approach with the BERT-based approach and develop a system that can provide an automated prediction of whether a given news article is fake or real.

SOFTWARE REQUIREMENTS:

The project requires a computer system capable of running Python and the required Natural Language Processing, machine learning, and deep learning libraries.

Python will be used as the main programming language for data preprocessing, feature extraction, model development, training, and evaluation.

Jupyter Notebook or Google Colab can be used as the development environment. Libraries such as Pandas and NumPy will be used for handling and processing the dataset. Matplotlib and Seaborn will be used for data visualization and performance analysis. Scikit-learn will be used for dataset splitting, TF-IDF feature extraction, Logistic Regression, preprocessing, and evaluation metrics. NLTK or similar NLP libraries can be used for text preprocessing. The Hugging Face Transformers library can be used to implement the BERT Transformer model.

TF-IDF will be used as the primary text feature extraction technique, and Logistic Regression will be used as the primary machine learning classifier. A pre-trained BERT model will be used as an advanced approach for comparison.

Since BERT can require considerable computational resources during training, Google Colab with GPU support is recommended for the BERT-based experiments. A standard computer can be used for preprocessing, TF-IDF processing, and Logistic Regression training.

CATEGORY REQUIREMENTS:

CATEGORY	REQUIREMENT
Operating System	Windows 10 / Windows 11
Programming Language	Python 3.x
Development Environment	Jupyter Notebook / Google COlab
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, NLTK, Transformers
Processor	Intel Core i3 or above
RAM	Minimum 8 GB
Storage	Minimum 5 GB free space
Dataset	Fake News Detection Dataset
Classification	Fake News / Real News
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, Confusion Matrix
GPU	Recommended for BERT training
NLP Techniques	Text Cleaning, Tokenization, Stop-word Removal, Text Preprocessing
Feature Extraction	TF-IDF
Advanced Model	BERT

Signature of the Guide

Course Instructor